

MI RESEARCH PACKAGE — SUPPLEMENTARY SLIDES

AI-Assisted Research Sprint · April 2026

Ana Clara Zoppi Serpa

CONTENTS

1. The Full Taxonomy of MI Techniques
 - 8 Families of MI Techniques
 - Activation Analysis & Attention Patterns
 - Circuit Discovery
 - Automated Interpretability & Probing
 - Similarity, Dynamics & Behavioral Testing
2. Injection Attacks via Creative Writing
 - The Creative Writing Attack Class

- MI for Understanding Jailbreaks — The Frontier

3. The Poetry Lens Experiment

- Experiment Design
- Poetry Lens — Task Deviation Results
- Poetry Lens — Deviation Profiles
- Poetry Lens — Task Override Heads

4. Ten MI Experiments on Prompt Injection

- Exp 1: The Tipping Point
- Exp 2: Injection Inoculation
- Exp 6: The Language Barrier
- Exp 8: Activation Steering for Defense
- Exp 10: The Confidence Paradox

5. Key Takeaways

- Cross-Experiment Synthesis
- Tooling Ecosystem

THE FULL TAXONOMY OF MI TECHNIQUES

Beyond Vectors & Words

MI is not just Logit Lens and SAEs — there are at least **8 distinct families**, each computing very different things.

1. Activation Analysis
2. Attention Pattern Analysis
3. Circuit Discovery
4. Automated Interpretability
5. Probing & Concept-Based Methods
6. Representation Similarity
7. Training Dynamics
8. Behavioral & Black-Box Testing

1. Activation Analysis

- Logit Lens / Tuned Lens
- Sparse Autoencoders (SAEs)
- Activation Steering
- Probing Classifiers

Output: token rankings, feature activations

2. Attention Patterns

- Attention head visualization
- Attention knockout / ablation
- Induction head detection

Output: heatmaps, matrices, graph patterns

3. The Crown Jewel of MI

Finding the **computational graph** (subnetwork) responsible for a specific behavior.

- **Manual circuit analysis:** IOI circuit — 26 heads in 7 functional classes
- **ACDC:** Automatically finds circuits by pruning edges (NeurIPS 2023)
- **Sparse Feature Circuits:** SAE features as nodes — monosemantic (ICLR 2025)
- **Attribution Graphs:** Anthropic's 2025 method — traced circuits in Claude 3.5 Haiku

A circuit is a *program* or *algorithm* implemented by a subset of the network.

4. Let AI Explain AI

- Neuron-to-text (GPT-4 labeling GPT-2 neurons)
- Neuronpedia — interactive feature dashboards
- Automated auditing agenda (Oxford, 2026)

5. Does the Model Know X?

- Linear probes for concepts
- Concept erasure (LEACE, INLP)
- Truth probes — models have internal truthfulness!

6. Similarity

- CKA, CCA, RSA
- Model Stitching

Compare
models

7. Dynamics

- Phase transitions
- Loss landscape
- Lottery tickets

How
models
learn

8. Black Box

- Contrast pairs
- Causal abstraction
- Influence functions

Input →
Output
tests

INJECTION ATTACKS VIA CREATIVE WRITING

Key Insight

Safety training is optimized against **prose**. Meter, rhyme, and metaphor push inputs **out-of-distribution**.

- **Adversarial Poetry** (Prandi et al., 2025): 62% ASR across 25 models — poetry is **18× more effective** than prose
- **Adversarial Tales** (Bisconti et al., 2026): 71.3% ASR via cyberpunk narratives
- **Doublespeak** (Yona et al., 2025): “carrot” converges to “bomb” layer by layer — 74% ASR
- **Jailbreak Mimicry** (Ntais, 2025): auto-generated narrative jailbreaks — 81% ASR

- **Attention Slipping** (ICLR 2026): Model gradually reduces attention to unsafe content during attacks
- **Safety Knowledge Neurons** (EACL 2026): Specific neurons control refusal — >97% effectiveness
- **Jailbreak Probes** (BlackboxNLP 2025): Different jailbreaks use distinct internal mechanisms
- **Circuit Discovery for Jailbreak Detection**: Ablating jailbreak circuits reduces ASR by up to 80%
- **Geometry of Refusal** (ICML 2025): Refusal is mediated by multiple independent directions, not one

These are the building blocks for your research contribution.

THE POETRY LENS EXPERIMENT

Research Question

When an LLM is primed for a specific task, what happens **internally** when a user injects a creative-writing-framed instruction?

- **Model:** GPT-2 Small (124M params, 12 layers, CPU)
- **Task:** English → French translation via 3-shot few-shot context
- **Prompts:** 12 total — 3 normal, 3 prose injection, 3 poetic, 3 narrative

H1: Which framing causes the strongest task deviation?

Measured by average $P(\text{French})$ at the final layer

- Normal (baseline): **0.1869**
- Prose injection: **0.1374** (-26.5%)
- Poetic injection: **0.1470** (-21.3%)
- Narrative injection: **0.1521** (-18.6%)

Result: **Prose wins, not poetry!** On GPT-2 Small (no instruction tuning), direct commands cause the strongest deviation. The poetry advantage likely requires instruction-tuned models.

H2: Deviation trajectories across layers

Late/Early deviation ratio — lower = more uniform deviation

- Normal: 4.9× late/early ratio
- Prose: **5.4×** (steep late divergence)
- Poetry: **1.8×** (most uniform across layers!)
- Narrative: **2.1×**

Poetry deviation is **evenly distributed** from early to late layers. Creative writing produces representations that are *immediately* different.

Specific heads that respond to injection

Dramatic attention shifts from normal → injection processing

- **L7H5:** 4.2% → 50.9% attention to input (+46.7pp) — “context switch” detector
- **L0H6:** 12.3% → 55.9% (+43.6pp) — early-layer novel input responder
- **L9H2:** 14.2% → 54.4% (+40.2pp) — late-layer injection content locker

L5H8 *resists* injection — maintains few-shot attention. A “task loyalty” head.

TEN MI EXPERIMENTS ON PROMPT INJECTION

At which token does the model commit to deviating?

Feed injection prompts one token at a time, measure $P(\text{French})$.

- Prose: tips at token 2 (“Ignore **the**”)
- Poetry: tips at **token 0** (“For”)
- Narrative: tips at **token 0** (“Once”)

Creative writing deviates the model from the **very first token**. Unusual opening tokens are themselves a signal for task deviation.

Can clamping injection-sensitive attention heads prevent deviation?

Clamped top 5 heads (L0H6, L9H0, L9H2, L10H4, L7H5) to baseline values.

Result: Zero effect.

The injection mechanism is **NOT mediated by attention**. Deviation happens in the **residual stream / MLP pathway**. Attention heads are *correlates*, not *causes*.

Does the language of injection matter?

Same override instruction in 5 different languages.

- English: $P(\text{French}) = 0.1855$
- French: $P(\text{French}) = \mathbf{0.1507}$ (garbled French output)
- Spanish: $P(\text{French}) = 0.1858$ (hybrid output!)
- German: $P(\text{French}) = \mathbf{0.0998}$ — most disruptive!

German is the most effective despite being semantically irrelevant. Injection disrupts through **token distribution**, not meaning.

Can we cancel out injection by subtracting the deviation direction?

Subtract $\alpha \times \text{deviation_direction}$ from layers 8–11 during inference.

- $\alpha = 0$: $P(\text{French}) = 0.1855$ (no defense)
- $\alpha = 10$: $P(\text{French}) = \mathbf{0.2083}$ (+12.3%)
- $\alpha = 20$: $P(\text{French}) = \mathbf{0.2164}$ (+16.7%)

Proof-of-concept for mechanistic defense against prompt injection. Smooth, monotonic recovery — no phase transitions.

Is the model more or less confident during injection?

Entropy and top-k probability mass measurements.

- Normal: 7.88 bits entropy, 10.6% top-1 prob
- Prose injection: **8.83 bits**, 6.9% top-1 (-35%!)
- Poetry injection: **8.81 bits**, 7.7% top-1

Injection creates **confusion, not redirection**. The model is *torn* between the few-shot context and the injection content.

KEY TAKEAWAYS

**Internal deviation $\neq$
behavioral deviation.
MI can detect injection
attempts**

1. The **first token** matters enormously — OOD openers trigger immediate deviation
2. Attention is a **red herring** for defense — the real mechanism is in MLP/residual stream
3. More context = more resistance (**1-shot is 4–6× more vulnerable** than 3-shot)
4. **Activation steering is a viable defense** — subtracting deviation direction works
5. Constrained tasks resist injection better; open-ended tasks are vulnerable
6. Injection creates **confusion**, not confident redirection

Libraries

- **nnsight** — any PyTorch model, local or via NDIF
- **TransformerLens** — hooks at every activation
- **SAELens** — SAE training & pre-trained loading
- **circuit-tracer** — Anthropic's attribution graphs

Infrastructure

- **NDIF** — NSF-funded, free, 80+ models
- Llama-3.1-405B, GPT-J-6B, DeepSeek-R1, Qwen, Gemma...
- Same code: just change model name + `remote=True`